# Customer Shopping Behavior Analysis

## Project Summary

This project studies customer shopping behaviour using transactional data from 3,900 purchases across multiple product categories. The aim is to understand and derive insights into spending habits, customer segments, product preferences, and subscription patterns. These findings will help guide smarter business decisions and shape strategies for sales, marketing, and customer engagement.

## Dataset Summary

Count of rows: 3900

Count of columns: 18

Key features:

Customer demographics.

Purchase details

Shopping behavior.

This dataset has missing values.

## EDA using Python

First step for analysis is to understand the data, look for inconsistency, and notice patterns etc. by exploratory data analysis.

- Load the data: Imported Pandas and imported the dataset.
- EDA : Did initial exploration to understand the overall data by using df.info() and df.summary()

```
df.describe(include='all')
```

| | Customer ID | Age | Gender | Item Purchased | Category | Purchase Amount (USD) | Location | Size | Color | Season | Review Rating | Subscription Status | Shipping Type | Discount Applied | Promo Code Used |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **count** | 3900.000000 | 3900.000000 | 3900 | 3900 | 3900 | 3900.000000 | 3900 | 3900 | 3900 | 3900 | 3863.000000 | 3900 | 3900 | 3900 | 3900 |
| **unique** | NaN | NaN | 2 | 25 | 4 | NaN | 50 | 4 | 25 | 4 | NaN | 2 | 6 | 2 | 2 |
| **top** | NaN | NaN | Male | Blouse | Clothing | NaN | Montana | M | Olive | Spring | NaN | No | Free Shipping | No | No |
| **freq** | NaN | NaN | 2652 | 171 | 1737 | NaN | 96 | 1755 | 177 | 999 | NaN | 2847 | 675 | 2223 | 2223 |
| **mean** | 1950.500000 | 44.068462 | NaN | NaN | NaN | 59.764359 | NaN | NaN | NaN | NaN | 3.750065 | NaN | NaN | NaN | NaN |
| **std** | 1125.977353 | 15.207589 | NaN | NaN | NaN | 23.685392 | NaN | NaN | NaN | NaN | 0.716983 | NaN | NaN | NaN | NaN |
| **min** | 1.000000 | 18.000000 | NaN | NaN | NaN | 20.000000 | NaN | NaN | NaN | NaN | 2.500000 | NaN | NaN | NaN | NaN |
| **25%** | 975.750000 | 31.000000 | NaN | NaN | NaN | 39.000000 | NaN | NaN | NaN | NaN | 3.100000 | NaN | NaN | NaN | NaN |
| **50%** | 1950.500000 | 44.000000 | NaN | NaN | NaN | 60.000000 | NaN | NaN | NaN | NaN | 3.800000 | NaN | NaN | NaN | NaN |
| **75%** | 2925.250000 | 57.000000 | NaN | NaN | NaN | 81.000000 | NaN | NaN | NaN | NaN | 4.400000 | NaN | NaN | NaN | NaN |
| **max** | 3900.000000 | 70.000000 | NaN | NaN | NaN | 100.000000 | NaN | NaN | NaN | NaN | 5.000000 | NaN | NaN | NaN | NaN |

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Customer ID             3900 non-null   int64
 1   Age                     3900 non-null   int64
 2   Gender                  3900 non-null   object
 3   Item Purchased          3900 non-null   object
 4   Category                3900 non-null   object
 5   Purchase Amount (USD)   3900 non-null   int64
 6   Location                3900 non-null   object
 7   Size                    3900 non-null   object
 8   Color                   3900 non-null   object
 9   Season                  3900 non-null   object
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   object
 12  Shipping Type           3900 non-null   object
 13  Discount Applied        3900 non-null   object
 14  Promo Code Used         3900 non-null   object
 15  Previous Purchases      3900 non-null   int64
 16  Payment Method          3900 non-null   object
 17  Frequency of Purchases  3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

- Handling missing values: Checked for null values (*with df.isnull().sum()*) and imputed missing values in **Review Rating** column using *Lambda function*.
- Standardization: Renamed columns to snake case for better readability and documentation.
- Feature engineering:
  - Created **age_group** column to classify customers based on their age.
  - Created **purchase_frequency_days** from purchase data.
- Data consistency check: Considering the objective of this project **promo_code_used** column was not adding any value. Hence it has been dropped.
- After cleaning the dataset, I connected the python script to PostgreSQL and loaded the dataset for further analysis.

## Data Analysis using PostgreSQL (Business Transactions)

I have performed analysis in PostgreSQL to derive insights as per business requirement.

1. Compared total revenue generated by male vs. female and their percentage of contribution.

| | gender text | revenue numeric | pct_contribution numeric |
|---|---|---|---|
| 1 | Female | 75191 | 32.26 |
| 2 | Male | 157890 | 67.74 |

2. Identified the customers who used a discount but still spent more than the average purchase amount.

| | customer_id bigint | purchase_amount bigint |
|---|---|---|
| 1 | 2 | 64 |
| 2 | 3 | 73 |
| 3 | 4 | 90 |
| 4 | 7 | 85 |
| 5 | 9 | 97 |
| 6 | 12 | 68 |
| 7 | 13 | 72 |
| 8 | 16 | 81 |
| 9 | 20 | 90 |
| 10 | 22 | 62 |
| 11 | 24 | 88 |
| 12 | 29 | 94 |
| 13 | 32 | 79 |
| 14 | 33 | 67 |
| 15 | 35 | 91 |
| 16 | 37 | 69 |

3. Identified top products with the highest average review rating.

| | products text | average_review numeric |
|---|---|---|
| 1 | Gloves | 3.86 |
| 2 | Sandals | 3.84 |
| 3 | Boots | 3.82 |
| 4 | Hat | 3.80 |
| 5 | Skirt | 3.78 |

4. Compared the average purchase amounts between standard and express shipping to understand the efficiency of respective shipping type.

| | shipping_type text | avg_purchase_amount numeric |
|---|---|---|
| 1 | Standard | 58.46 |
| 2 | Express | 60.48 |

5. Checked if subscribed customers spend more? Compared average spend and total revenue between subscribers and non-subscribers to understand the efficiency of subscription offers.

| | subscription_status text | total_customers bigint | average_spend numeric | total_revenue numeric |
|---|---|---|---|---|
| 1 | No | 2847 | 59.87 | 170436 |
| 2 | Yes | 1053 | 59.49 | 62645 |

6. Identified top 5 products with the highest percentage of discounted purchases.

| | products text | discount_rate numeric |
|---|---|---|
| 1 | Hat | 50.00 |
| 2 | Sneakers | 49.00 |
| 3 | Coat | 49.00 |
| 4 | Sweater | 48.00 |
| 5 | Pants | 47.00 |

7. Classified customers into ‘New’, ‘Returning’, ‘Loyal’ segment based on purchase history to understand who all can be a good resource of revenue.

| | customer_segment text | total_customers bigint |
|---|---|---|
| 1 | Loyal | 3116 |
| 2 | Returning | 701 |
| 3 | New | 83 |

8. Top 3 products within each category.

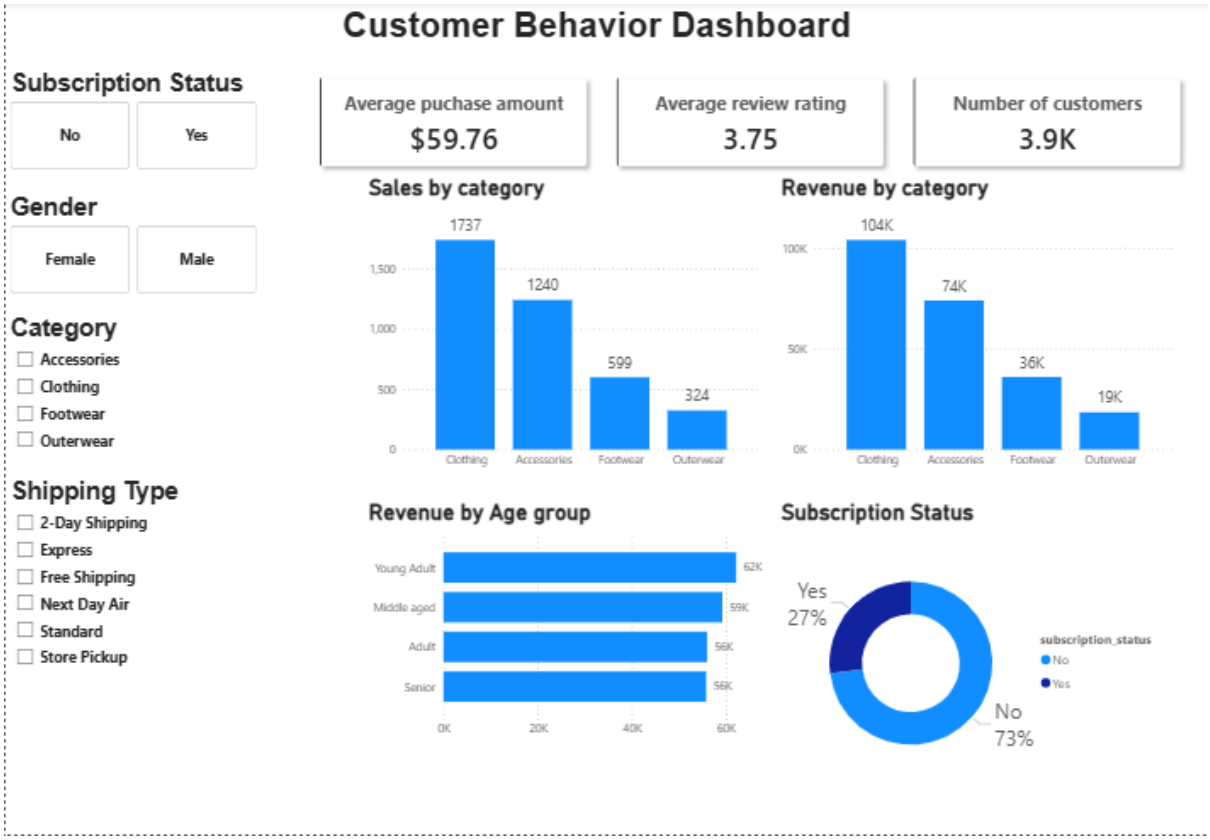
| | category text | item_purchased text | total_orders bigint | rnk bigint |
|---|---|---|---|---|
| 1 | Accessories | Belt | 161 | 2 |
| 2 | Accessories | Jewelry | 171 | 1 |
| 3 | Accessories | Scarf | 157 | 3 |
| 4 | Accessories | Sunglasses | 161 | 2 |
| 5 | Clothing | Blouse | 171 | 1 |
| 6 | Clothing | Dress | 166 | 3 |
| 7 | Clothing | Pants | 171 | 1 |
| 8 | Clothing | Shirt | 169 | 2 |
| 9 | Footwear | Sandals | 160 | 1 |
| 10 | Footwear | Shoes | 150 | 2 |
| 11 | Footwear | Sneakers | 145 | 3 |
| 12 | Outerwear | Coat | 161 | 2 |
| 13 | Outerwear | Jacket | 163 | 1 |

9. Wanted to understand if repeat buyers are limited to subscribed members only or not.

| | subscription_status text | total_customers bigint |
|---|---|---|
| 1 | No | 2518 |
| 2 | Yes | 958 |

10. Age group wise contribution toward revenue.

| | age_group text | total_purchase numeric | pct_contribution numeric |
|---|---|---|---|
| 1 | Young Adult | 62143 | 26.66 |
| 2 | Middle aged | 59197 | 25.40 |
| 3 | Adult | 55978 | 24.02 |
| 4 | Senior | 55763 | 23.92 |

## Dashboard in Power BI

After the analysis, to visually present the insights, I have used Power BI.